# EE3005 Image Processing (IP) — Assignment 2

*Image Enhancement, Linear and Non-Linear Filtering, and Edge Detection*

## Objective

The primary goal of this assignment is to develop a deeper understanding of image enhancement and spatial filtering techniques. Students will investigate adaptive histogram equalization, linear and non-linear filters, convolution and correlation, border padding, separable filtering, image sharpening, and edge detection.

Through practical implementation in MATLAB, students will learn how different filtering methods affect image quality, noise, fine details, and edge information. Particular emphasis will be placed on comparing methods under different image conditions, interpreting quantitative and visual results, and explaining the strengths and limitations of each approach.

Students are expected to implement image processing techniques, analyze results critically, and select appropriate methods for specific image processing problems.

## Task 1: Adaptive Histogram Equalization and CLAHE

### 1.1 Image Selection and Initial Analysis

- Select a grayscale biomedical image with limited contrast or uneven illumination.
- Display the original image and its grayscale histogram.
- Explain the main contrast-related limitations observed in the image.

### 1.2 Adaptive Histogram Equalization (AHE)

- Apply adaptive histogram equalization to the selected image.
- Display the processed image and its histogram.
- Explain how local histogram equalization differs from global histogram equalization.
- Identify regions where local contrast enhancement is particularly noticeable.

### 1.3 Contrast Limited Adaptive Histogram Equalization (CLAHE)

- Apply CLAHE to the same image.
- Investigate the effect of changing the clip limit and the size of the local regions.
- Display results obtained with at least two different parameter configurations.
- Explain how these parameters affect local contrast enhancement and noise amplification.

### 1.4 Comparative Analysis

- Compare global histogram equalization, AHE, and CLAHE.
- Identify which method provides the most useful enhancement for the selected image.
- Discuss whether stronger contrast enhancement necessarily produces a better image.

Support your conclusions with processed images, histograms, and observations specific to your selected image.

## Task 2: Convolution, Correlation, and Border Effects

### 2.1 Kernel Operations

Use the image matrix and filter kernel shown in Figure 1(a).

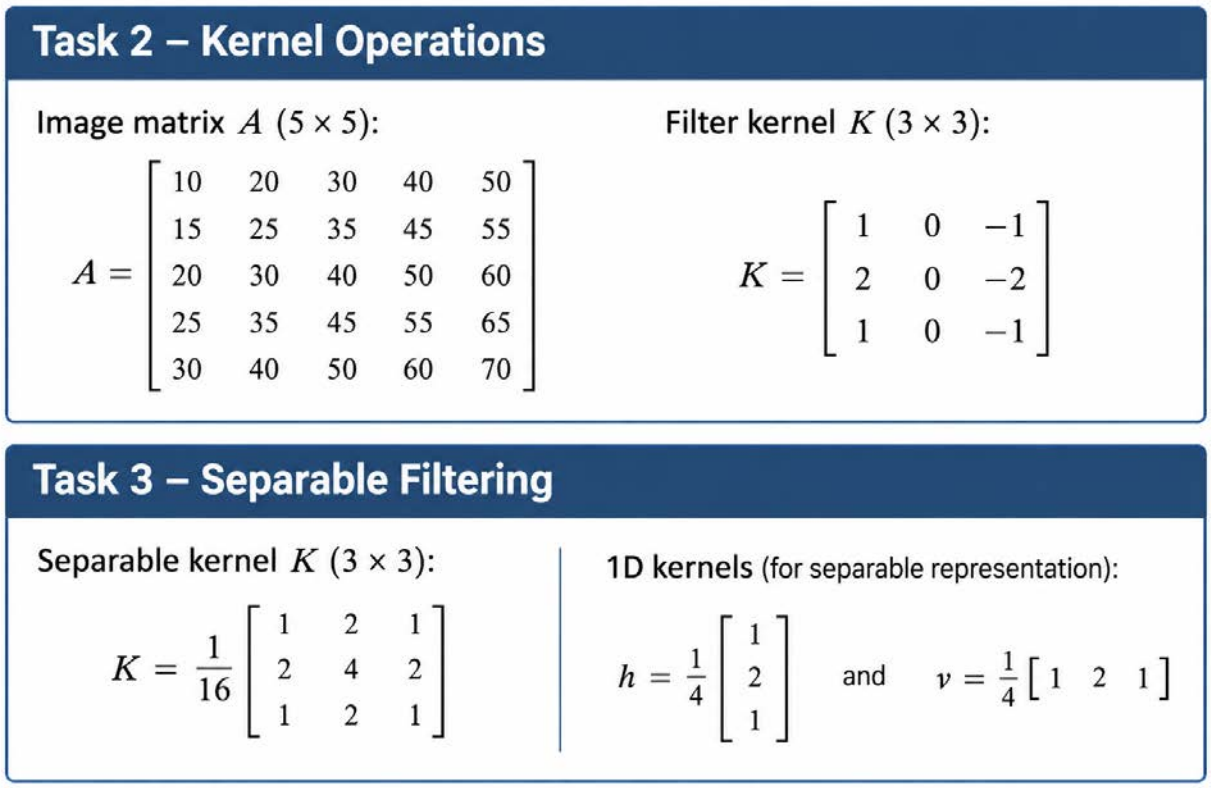



*Figure 1. Matrices for Task 2(a) and Task 3(b).*

- Explain the purpose of a kernel in spatial filtering.
- Calculate the output value at the center pixel using correlation.
- Calculate the corresponding output value using convolution. Flip the kernel horizontally and vertically before applying it.
- Show the intermediate element-wise multiplications and summation for both operations.
- Explain why the two results differ for this kernel.

### 2.2 Border Padding

- Apply a 3×3 arithmetic mean filter to a grayscale image using zero padding, replication (clamp) padding, reflection (mirror) padding, and circular (wrap) padding.
- Display the resulting images side by side.
- Focus your analysis on image boundaries and explain how the padding method affects the output.

### 2.3 Discussion

- Explain why border effects occur during filtering.
- Identify which padding method produces the most reasonable result for your selected image and justify your choice.
- Discuss whether the same padding method is necessarily optimal for every image and filter.

## Task 3: Separable Filtering and Computational Efficiency

### 3.1 Separable Kernel

Use the separable Gaussian kernel and its one-dimensional factors shown in the lower section of Figure 1(b).

- Determine whether the kernel is separable.
- Express the kernel as the outer product of two one-dimensional vectors.
- Verify mathematically that the resulting matrix is identical to the original kernel.

### 3.2 Implementation

- Apply the two-dimensional kernel directly to a grayscale image.
- Implement the equivalent operation using successive horizontal and vertical one-dimensional filtering.
- Display the two output images side by side.
- Calculate the maximum absolute pixel-wise difference between the two results.

### 3.3 Computational Analysis

- Compare the number of multiplication operations required per pixel for the two implementations, assuming a direct K×K filter and two one-dimensional filters of length K.
- Explain how separability can reduce computational cost.
- Discuss whether the two approaches should produce identical results and identify possible reasons for numerical or implementation differences.

## Task 4: Linear Filtering for Noise Reduction

### 4.1 Noise Generation

- Select a grayscale biomedical image and retain the original as a reference.
- Generate two noisy versions: one corrupted by Gaussian noise and another corrupted by salt-and-pepper noise.
- Display the original and noisy images.
- Explain the visible characteristics of the two noise types.

### 4.2 Mean and Gaussian Filtering

- Apply an arithmetic mean filter and a Gaussian filter to the noisy images.
- Use at least two different filter sizes or Gaussian parameter configurations.
- Display and compare the filtered results.
- Explain how smoothing affects noise, edges, and fine image details.

### 4.3 Quantitative Evaluation

- Calculate PSNR between the clean reference image and each noisy or filtered version.
- Present the results in a table.
- Identify the filter that achieves the highest PSNR for each noise type.

### 4.4 Critical Discussion

- Explain why a filter that reduces noise may also blur important structures.
- Discuss whether the filter with the highest PSNR necessarily produces the most useful image for visual inspection or medical interpretation.

Use the same reference image for all comparisons and ensure that all images have compatible dimensions and intensity ranges.

# Task 5: Non-Linear Filtering and Impulse Noise Removal

### 5.1 Median Filtering

- Apply median filtering to the salt-and-pepper noisy image from Task 4.
- Investigate at least two different neighborhood sizes.
- Compare the results with those obtained using the arithmetic mean filter.
- Explain why median filtering can be effective against impulse noise while preserving certain edges.

### 5.2 Mean Filter Variants

Investigate the geometric mean, harmonic mean, and contra-harmonic mean filters.

- Explain the mathematical principle of each filter.
- Apply each filter to an appropriate noisy image.
- Display each output and compare it with the original and noisy images.
- Discuss its effectiveness and limitations.

### 5.3 Contra-Harmonic Mean Filter Investigation

- Apply the contra-harmonic mean filter using at least one positive and one negative value of the order parameter Q.
- Test the filter on images corrupted separately by salt noise and pepper noise.
- Explain how changing the sign of Q affects suppression of these noise types.
- Identify cases where the filter fails or produces undesirable results.

### 5.4 Comparative Evaluation

- Calculate PSNR for filtered images whenever a clean reference is available.
- Summarize the results in a comparison table.
- Recommend a suitable filter for each noise type and justify your recommendations using your experimental results.

# Task 6: Image Sharpening Using Unsharp Masking

### 6.1 Unsharp Masking Implementation

Select a grayscale image containing visible edges and fine details. Apply a smoothing filter to obtain a blurred version of the original image.

$$m(x, y) = f(x, y) - f_{blur}(x, y)$$

$$g(x, y) = f(x, y) + \gamma\, m(x, y)$$

Here, $\gamma$ is the sharpening gain parameter.

### 6.2 Parameter Investigation

- Apply unsharp masking using at least three different values of $\gamma$.
- Display the original, blurred, and sharpened images.
- Explain how increasing the sharpening gain affects edges, fine details, and noise.

### 6.3 Critical Analysis

- Identify a parameter configuration that improves the visibility of details without producing excessive artifacts.
- Explain why excessive sharpening may be undesirable.

- Compare unsharp masking with direct high-frequency detail enhancement, commonly described as sharp masking in the lecture material.

## Task 7: Edge Detection Using Sobel, Laplacian, LoG, and DoG

### 7.1 First- and Second-Order Derivatives

- Apply Sobel filtering to investigate intensity changes and edge information.
- Apply a Laplacian operator to investigate second-order intensity changes.
- Display the results and explain the differences between first-order and second-order derivative-based approaches.

### 7.2 Laplacian of Gaussian (LoG)

- Apply Gaussian smoothing before computing the Laplacian response.
- Display the resulting response and identify prominent edge structures.
- Explain why Gaussian smoothing is used before second-order derivative-based edge detection.

### 7.3 Difference of Gaussian (DoG)

- Apply two Gaussian filters with different standard deviations to the same image.
- Subtract the two smoothed images to obtain a DoG response.
- Investigate how changing the Gaussian standard deviations affects the resulting structures.
- Explain the relationship between DoG and LoG.

### 7.4 Comparative Analysis

- Compare Sobel, Laplacian, LoG, and DoG results.
- Discuss their sensitivity to noise, ability to highlight boundaries, and computational considerations.
- Identify which method is most appropriate for the selected image and justify your conclusion.

Discuss specific structures or edges that are enhanced, missed, or affected by noise rather than relying solely on visual impressions.

## Task 8: Integrated Image Processing Pipeline

### 8.1 Design a Processing Pipeline

Design a processing pipeline for a grayscale biomedical image containing noise, limited contrast, and important structural boundaries.

- Include one contrast enhancement method from Task 1.
- Include one noise reduction method from Task 4 or Task 5.
- Include one sharpening or edge detection method from Task 6 or Task 7.

### 8.2 Implementation and Visualization

- Implement the complete pipeline in MATLAB.
- Display the original image and the intermediate result after each processing stage.
- Explain the purpose of each selected operation and the order in which the operations are applied.

### 8.3 Evaluation and Justification

- Compare the final result with the original image.
- Evaluate whether the pipeline improves contrast, reduces noise, and preserves important details.
- Identify at least one limitation of your pipeline.

- Propose one alternative processing sequence and explain why it might produce a different result.

Your final discussion must be based on the actual output of your implementation rather than a general description of the methods.

## Submission Guidelines

- One submission per group is sufficient.
- Submit the entire report as a single Word document.
- Include all MATLAB source code, original and processed images, relevant histograms, quantitative results, and explanations.
- Clearly label all figures and tables and refer to them in the report.
- Include the source of each image and briefly describe any noise added artificially.
- Explain the parameter values used in each experiment and justify important choices.
- Present intermediate calculations where explicitly requested, especially in Task 2 and Task 3.
- Include observations, limitations, unexpected results, and challenges encountered during implementation.
- Your report must reflect your own implementation and analysis. Be prepared to explain your code, calculations, parameter choices, and conclusions if clarification is requested.
- Generic explanations that are not supported by your own results will not be sufficient for full credit.

## Grading Rubric — 100 Points

| Criteria | Description | Points |
|---|---|---|
| 1. Adaptive Histogram Equalization and CLAHE | Correct implementation, parameter investigation, visualization, and comparison of global HE, AHE, and CLAHE. | 15 |
| 2. Convolution, Correlation, and Border Effects | Correct manual calculations, kernel flipping, padding implementations, and boundary analysis. | 15 |
| 3. Separable Filtering | Correct separability verification, implementation, output comparison, and computational analysis. | 10 |
| 4. Linear Filtering and Noise Reduction | Appropriate noise generation, mean/Gaussian filtering, PSNR calculation, and critical interpretation. | 10 |
| 5. Non-Linear Filtering | Implementation and comparison of median, geometric, harmonic, and contra-harmonic filters; correct interpretation of noise suppression. | 20 |
| 6. Unsharp Masking | Correct implementation, parameter investigation, sharpening analysis, and discussion of artifacts. | 10 |
| 7. Edge Detection | Correct implementation and meaningful comparison of Sobel, Laplacian, LoG, and DoG. | 15 |
| 8. Integrated Pipeline | Coherent pipeline design, intermediate results, justification of operation order, and limitation analysis. | 5 |
| TOTAL | | 100 |